



H2S

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Team Name : [YOUR TEAM NAME]

Team Leader Name : [YOUR NAME]

Problem Statement : AI-enabled Detection of Exoplanets from Noisy Astronomical Light Curves

LIVE DEMO <https://apex-oracle-production.up.railway.app>

Team Members

Team Leader:

Name:

College:

Team Member-1:

Name:

College:

Team Member-2:

Name:

College:

Team Member-3:

Name:

College:

Opportunity and USP

- Most existing entries are black-box detectors that only flag a dip in brightness. Ours also says what the signal is, measures it, and shows its reasoning.
- It works end to end: noise-robust detrending keeps faint transits, a classifier plus a centroid check tells real planets from look-alike binaries, and a physics model fits the period, depth, and duration.
- USP: detection, classification, parameters with confidence, and clear visual explanations, all in one pipeline you can actually inspect.
- Already validated on real TESS data: an AI model trained on real light curves beats classical vetting 0.95 vs 0.75 accuracy in 5-fold cross-validation (65 targets), flagging 90% of real false positives while missing zero planets.

Features of the solution

- Transit-safe detrending (Gaussian Process and Savitzky-Golay) that removes noise but keeps the 100 ppm dips.
- Four-way classification: planet transit, eclipsing binary, blend, or starspot.
- Learns the U-shape versus V-shape difference using attention.
- Estimates period, depth, and duration with calibrated error bars.
- Rejects blended false positives using centroid and difference imaging.
- Recovers single-transit, long-period planets that folding methods miss.
- Phase-folded plots with attention overlays that explain each decision.
- Honest validation through injection-recovery tests and SNR.

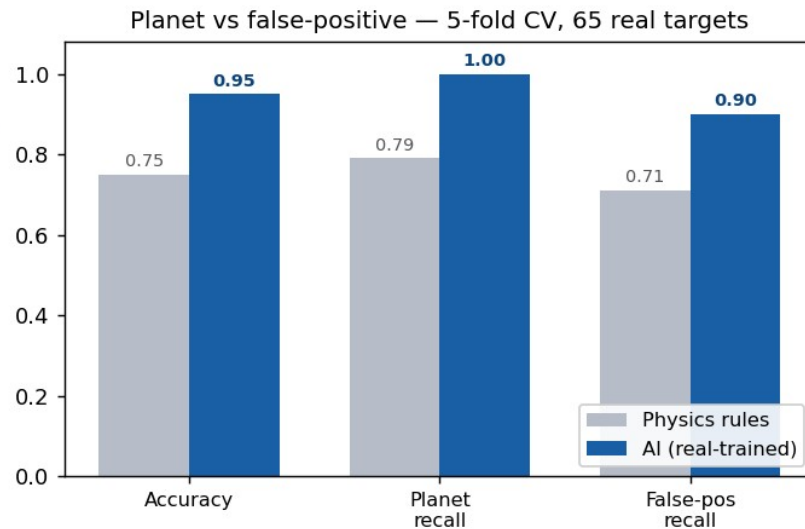
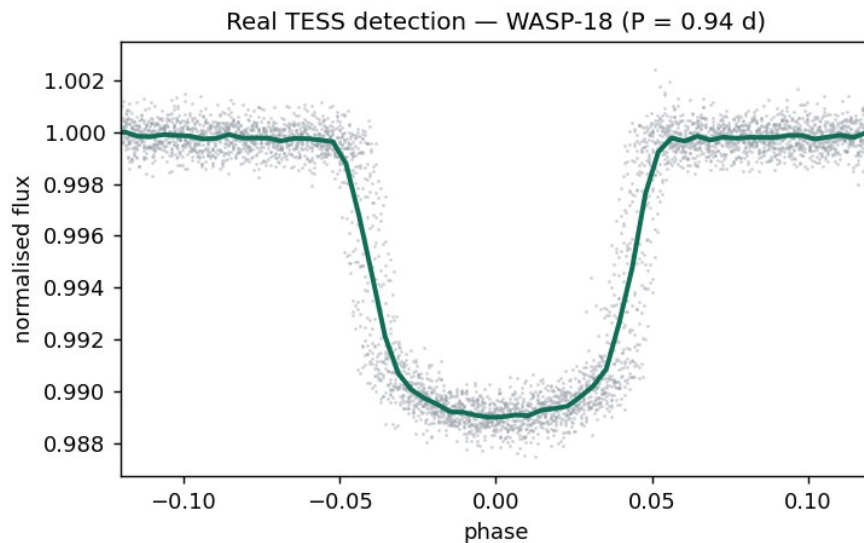
Process flow

- Nine stages: ingestion, detrending, candidate search and phase fold, classification, parameter estimation, vetting, explainability, validation, and a live demo.
- Ingestion pulls TESS light curves and labels with lightkurve and astroquery.
- A Gaussian Process detrends, then transitleastsquares finds candidates and folds them.
- The folded signal is classified, fitted with a physics model, and checked for false positives.
- The output is a class, parameters, a confidence level, and explanatory plots in a dashboard.

A full process-flow diagram can be added on this slide.

Validation on real TESS data

- AI trained on real TESS light curves beats classical vetting: 0.95 vs 0.75 accuracy (5-fold CV, 65 catalogue targets), and recovers real planets like WASP-18b at $P=0.94$ d.



Architecture

- A hybrid of three parts: a fast candidate front-end, a Transformer classification spine, and a differentiable-physics parameter head.
- Data layer: TESS light curves and pixel data, Gaia neighbours, and TOI labels.
- Front-end: detrending and BLS/TLS produce clean transit candidates.
- Spine: a multimodal Transformer labels transit, eclipsing binary, blend, or starspot.
- Head: a differentiable batman model with simulation-based inference returns parameters and posteriors.
- Centroid and difference-imaging vetting removes false positives before the result is shown.
- Reproducibility layer: uv, Hydra, Weights and Biases, and PyTorch Lightning.

The architecture diagram can be added on this slide.

Technologies used

- Language and environment: Python 3.11 with uv.
- Data and astronomy: lightkurve, astropy, astroquery, transitleastsquares, wotan.
- Detrending and Gaussian Processes: wotan, celerite2, gpytorch.
- Deep learning: PyTorch, Lightning, timm, Transformers, XGBoost.
- Physics and Bayesian inference: batman, sbi, dynesty, emcee.
- Vetting and explainability: triceratops, attention maps, matplotlib, plotly.
- Tracking, config, and demo: Hydra, OmegaConf, Weights and Biases, Streamlit.

Estimated implementation cost (optional)

- Data: the TESS archive from MAST is free.
- Software: the entire stack is open-source and free.
- Compute: free-tier GPUs (Kaggle or Colab), or a short cloud-GPU rental at roughly Rs 0 to 8,000.
- Total: close to zero. The real cost is engineering effort, not money.

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THANK YOU